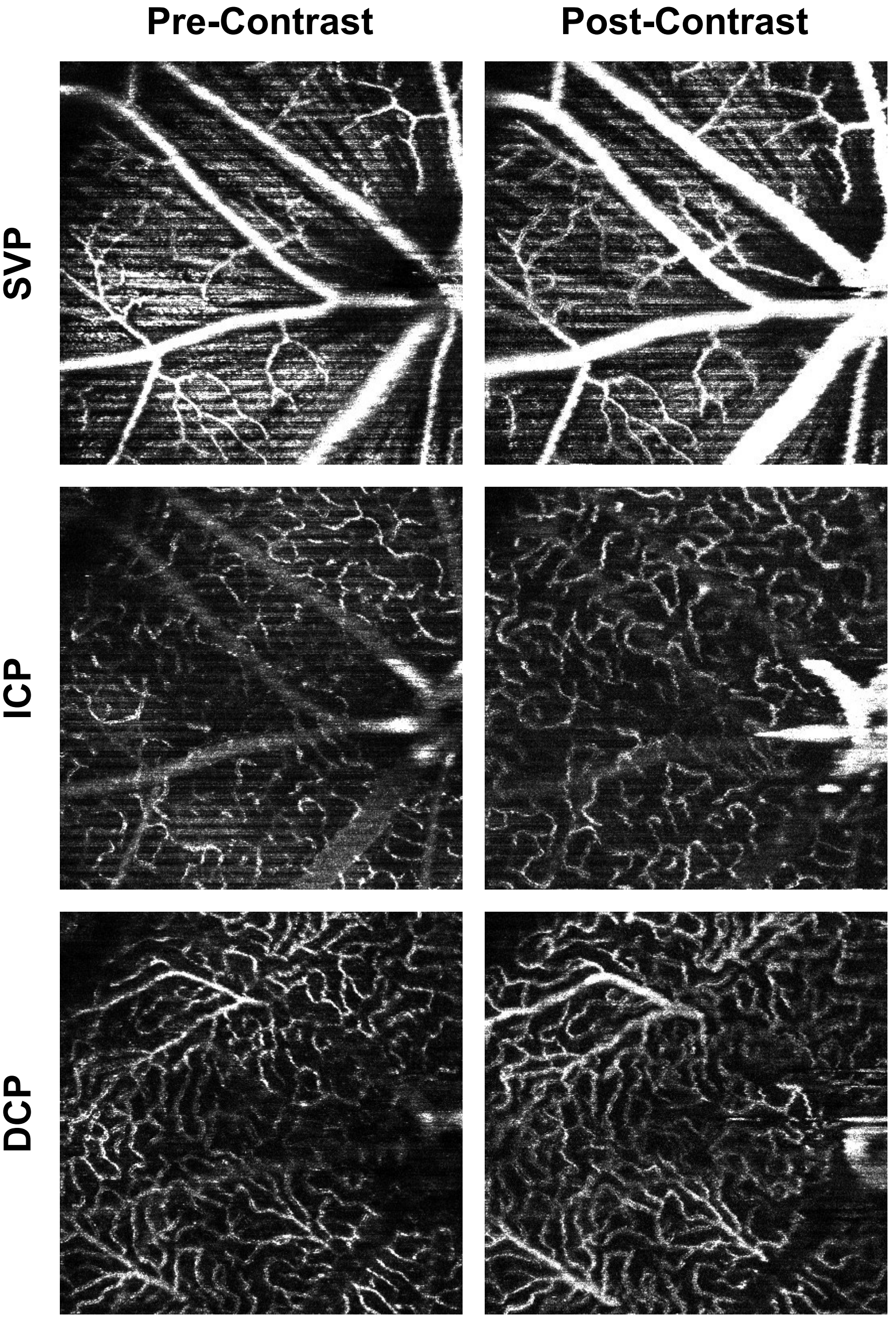
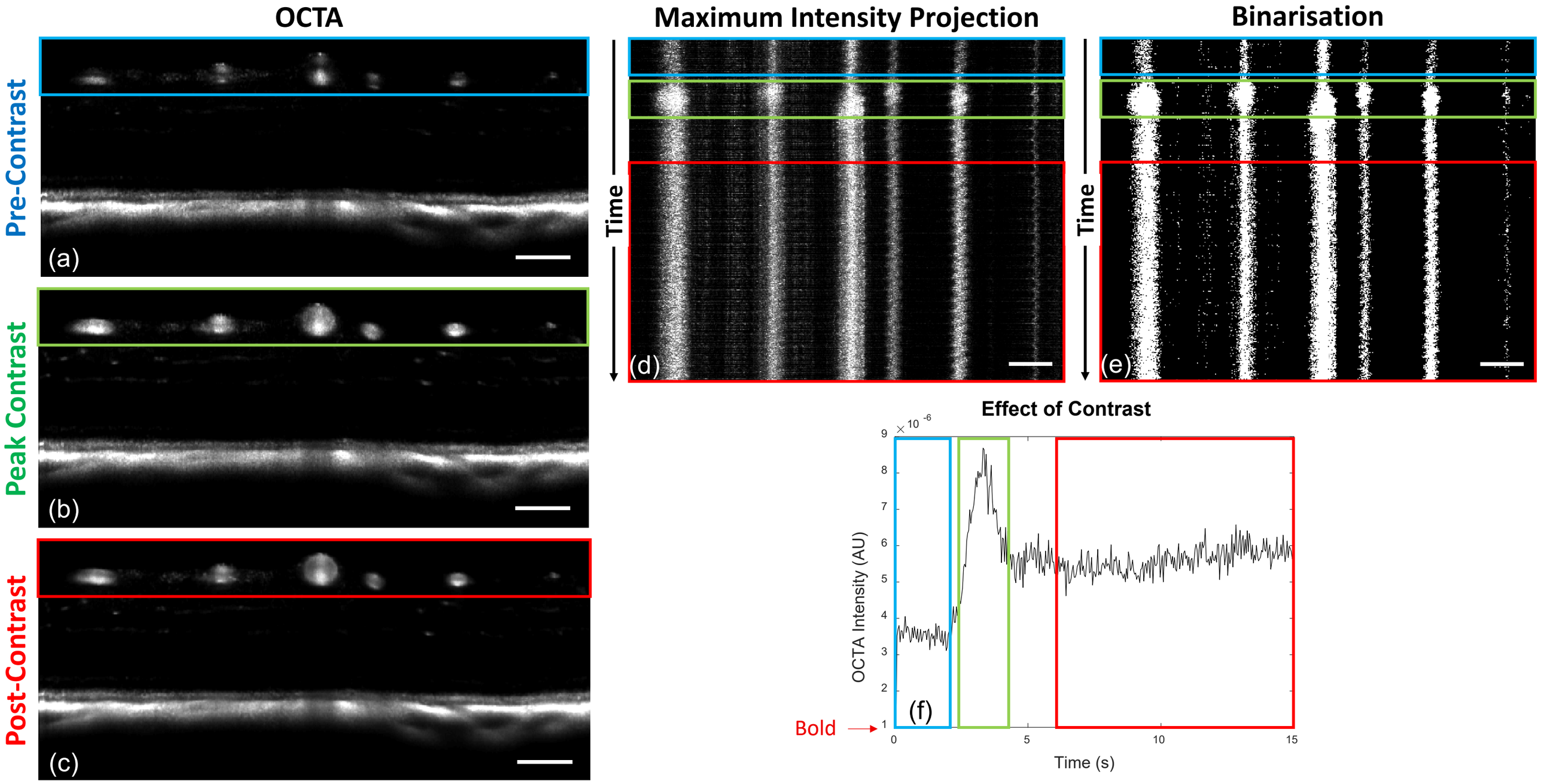
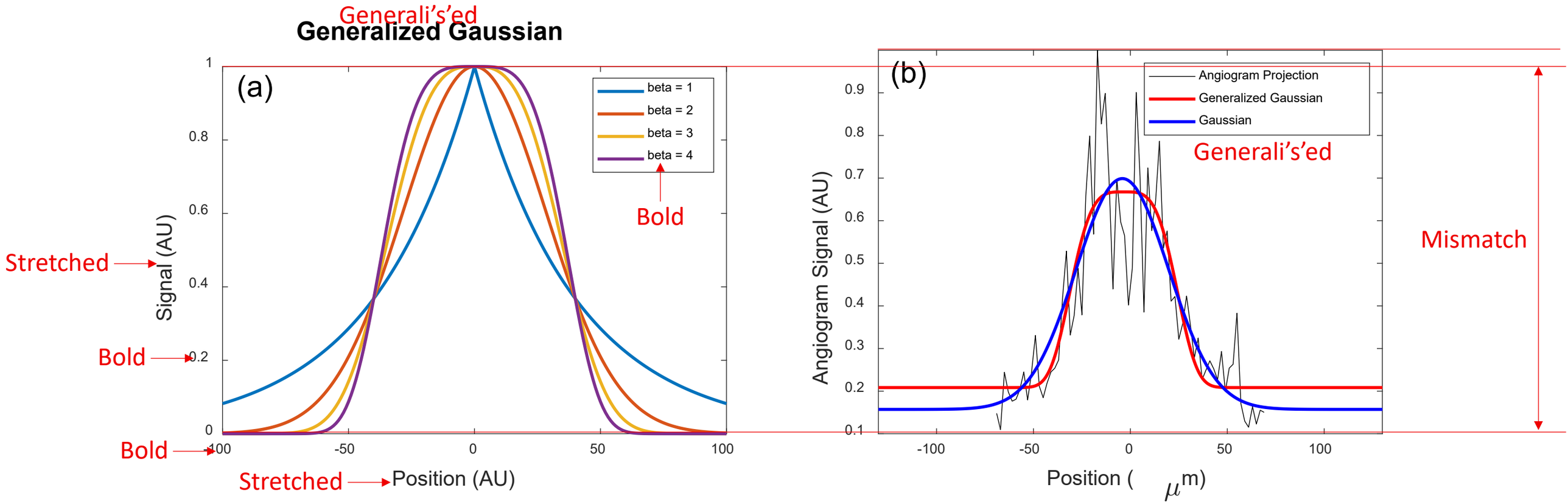
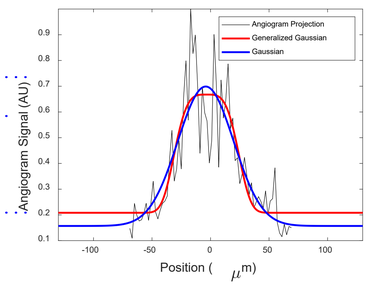

Pre-Contrast
Post-Contrast
SVP
ICP
DCP

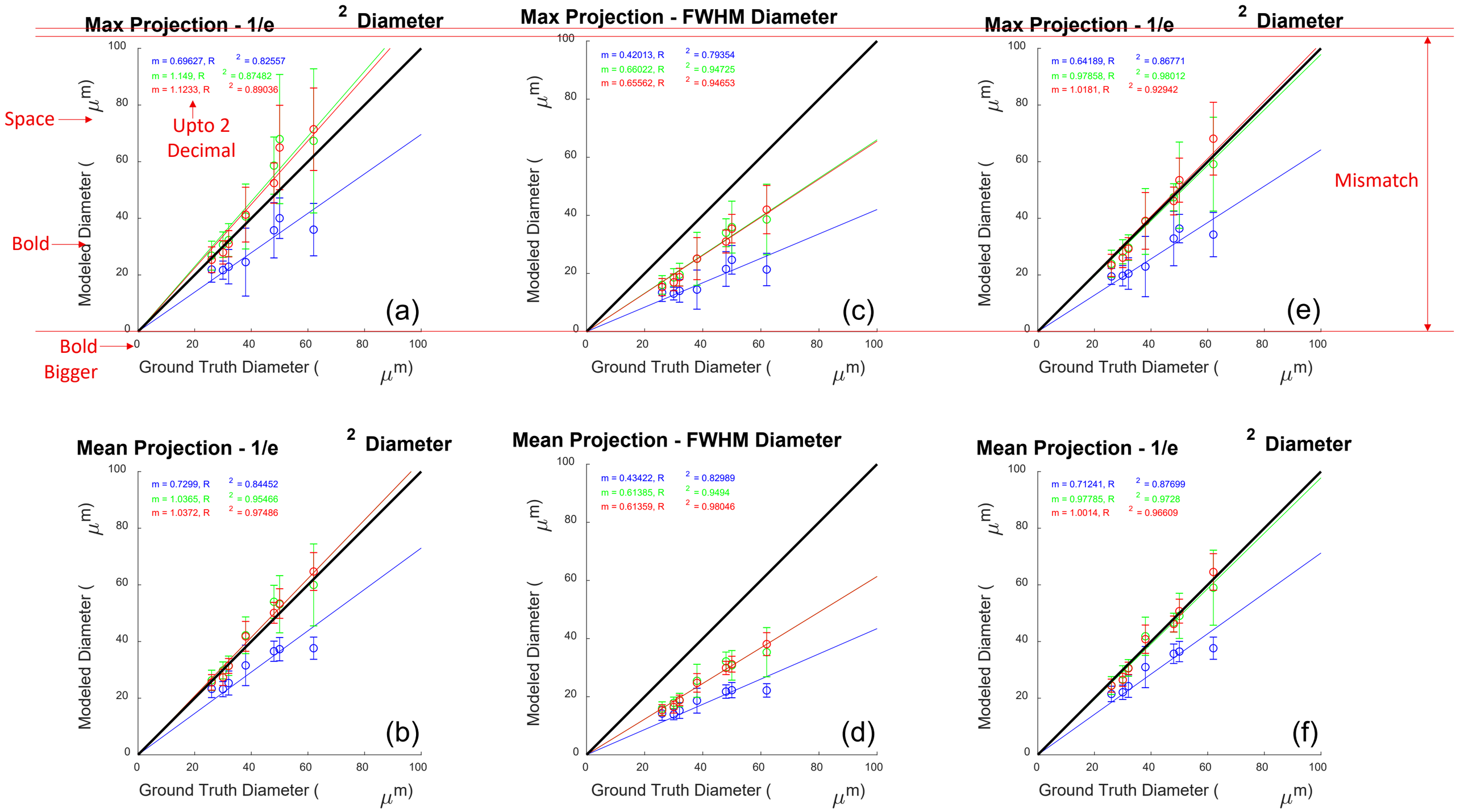

OCTA
Maximum Intensity Projection
Binarisation
Pre-Contrast
Peak Contrast
Post-Contrast
(a)
(b)
(c)
Time
(d)
Time
(e)
Effect of Contrast
$\times 10^{-6}$
OCTA Intensity (AU)
(f)
Bold
Time (s)
0
5
10
15
1
2
3
4
5
6
7
8
9

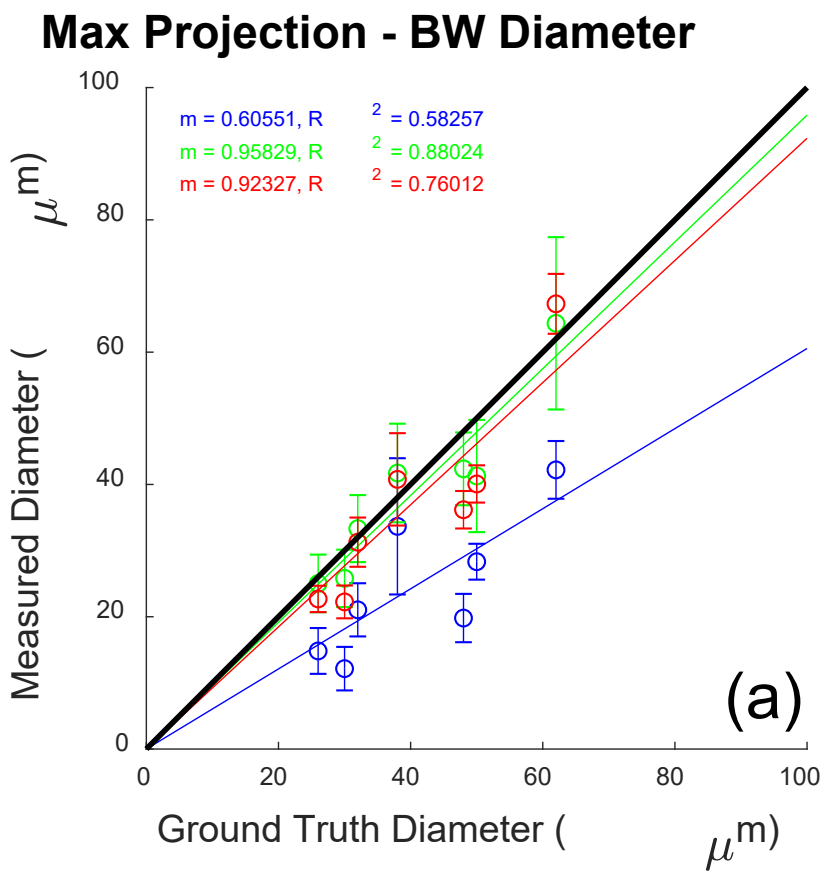

Generali's'ed
Generalized Gaussian
(a)
beta = 1
beta = 2
beta = 3
beta = 4
Bold
Stretched
Bold
Bold
Stretched
Signal (AU)
Position (AU)
(b)
Angiogram Projection
Generalized Gaussian
Gaussian
Generali's'ed
Mismatch
Angiogram Signal (AU)
Position (μm)

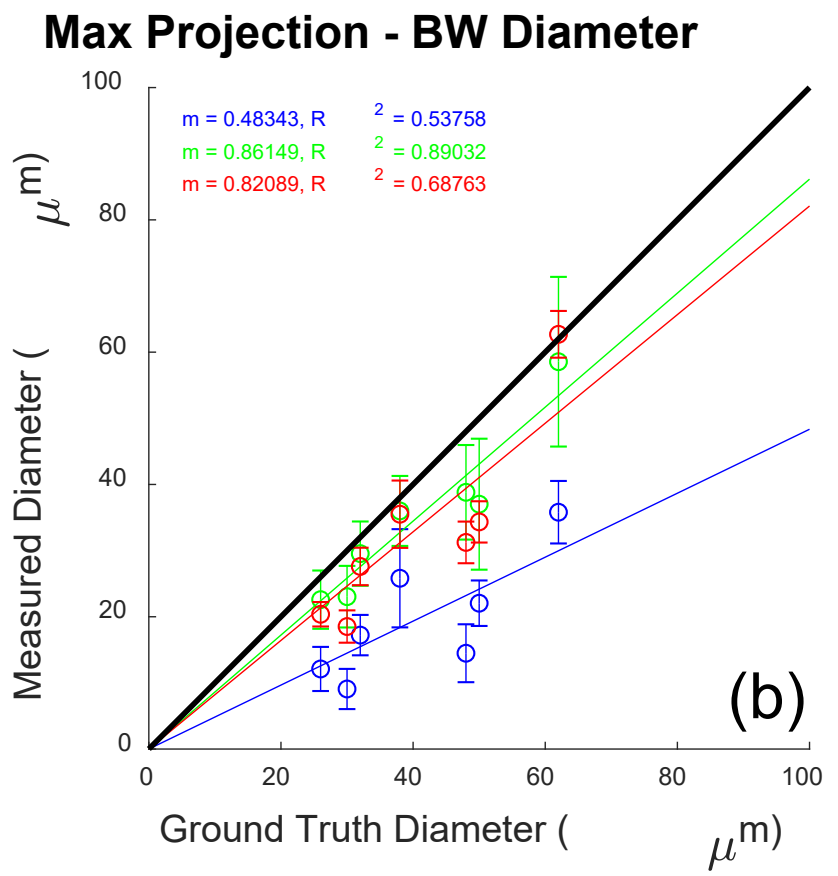
```
ft_gg = fittype( ...
    '(A.*exp(-(abs(x-mu)./alpha).^beta)+C)', ...
    'independent','x', ...
    'coeff', {'A','mu','alpha','beta','C'} ...
    );

% Normal Gaussian
ft_g = fittype( ...
    '(A.*exp(-(x-mu).^2./(2.*sigma.^2))+C)', ...
    'independent','x', ...
    'coeff', {'A','mu','sigma','C'} ...
    );
```

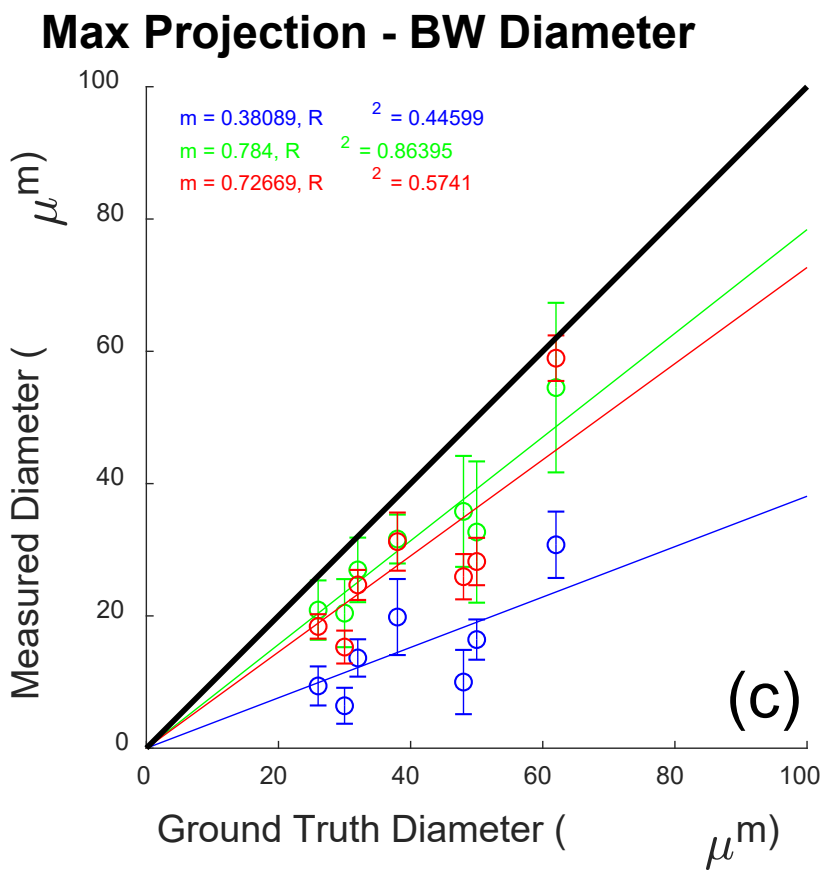


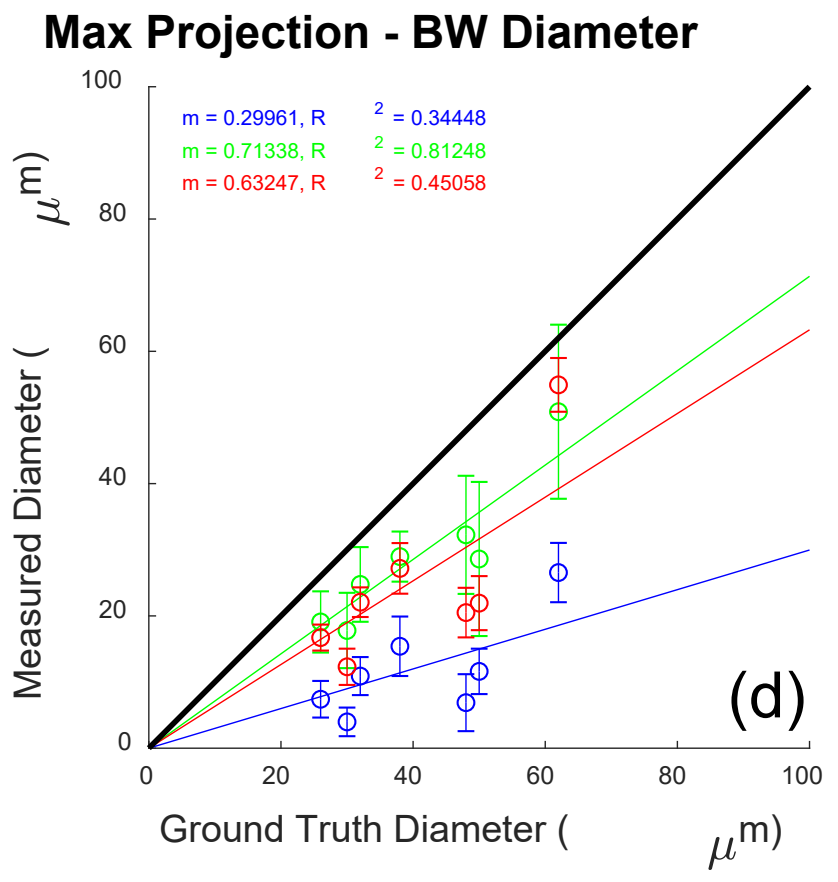
GG's extra parameter costs about 40% additional Computation Time

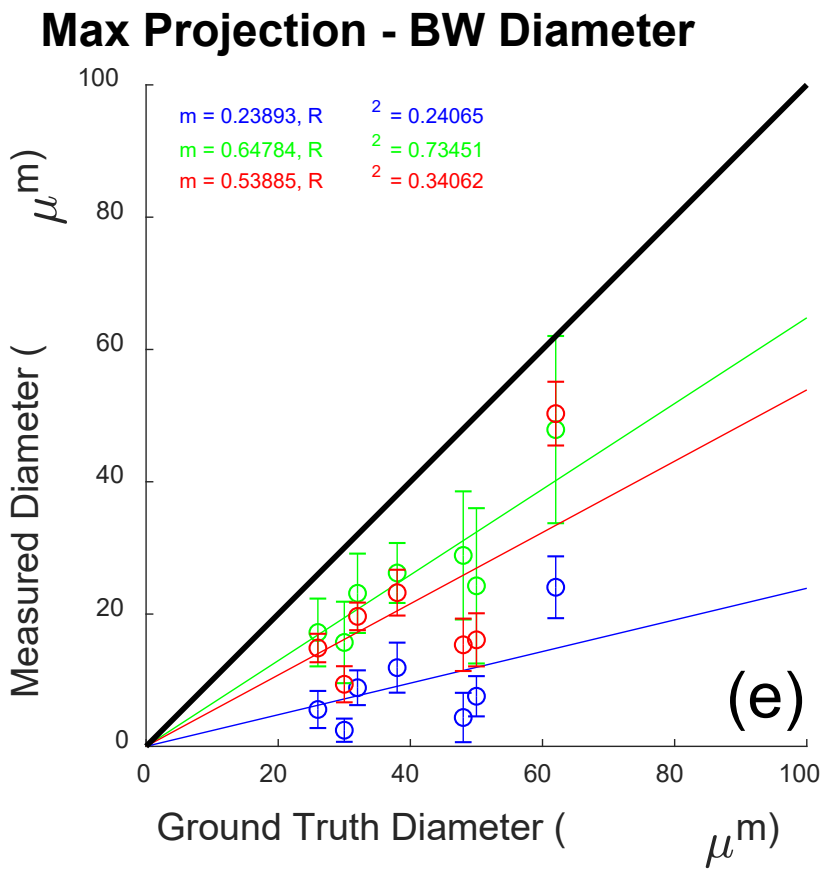

Max Projection - 1/e 2 Diameter
m = 0.69627, R 2 = 0.82557
m = 1.149, R 2 = 0.87482
m = 1.1233, R 2 = 0.89036
Space
Upto 2 Decimal
Bold
Bold Bigger
(a)
Max Projection - FWHM Diameter
m = 0.42013, R 2 = 0.79354
m = 0.66022, R 2 = 0.94725
m = 0.65562, R 2 = 0.94653
(c)
Max Projection - 1/e 2 Diameter
m = 0.64189, R 2 = 0.86771
m = 0.97858, R 2 = 0.98012
m = 1.0181, R 2 = 0.92942
Mismatch
(e)
Mean Projection - 1/e 2 Diameter
m = 0.7299, R 2 = 0.84452
m = 1.0365, R 2 = 0.95466
m = 1.0372, R 2 = 0.97486
(b)
Mean Projection - FWHM Diameter
m = 0.43422, R 2 = 0.82989
m = 0.61385, R 2 = 0.9494
m = 0.61359, R 2 = 0.98046
(d)
Mean Projection - 1/e 2 Diameter
m = 0.71241, R 2 = 0.87699
m = 0.97785, R 2 = 0.9728
m = 1.0014, R 2 = 0.96609
(f)
Modeled Diameter (μm)
Ground Truth Diameter (μm)
0
20
40
60
80
100

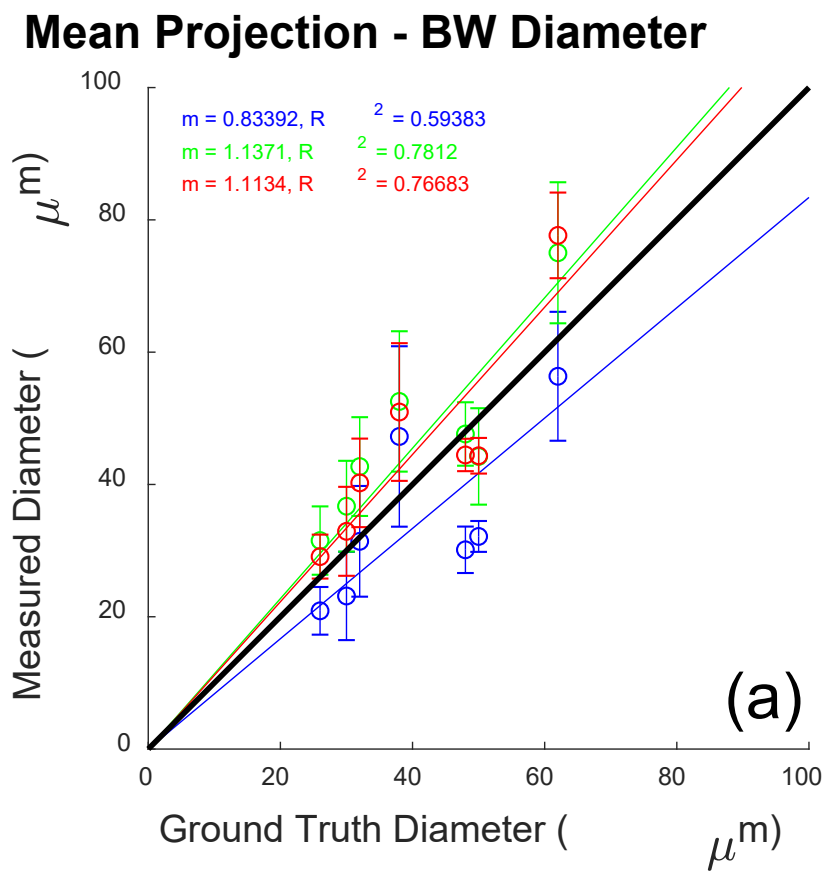

Max Projection - BW Diameter
m = 0.60551, R 2 = 0.58257
m = 0.95829, R 2 = 0.88024
m = 0.92327, R 2 = 0.76012
(a)
Measured Diameter (μm)
Ground Truth Diameter (μm)

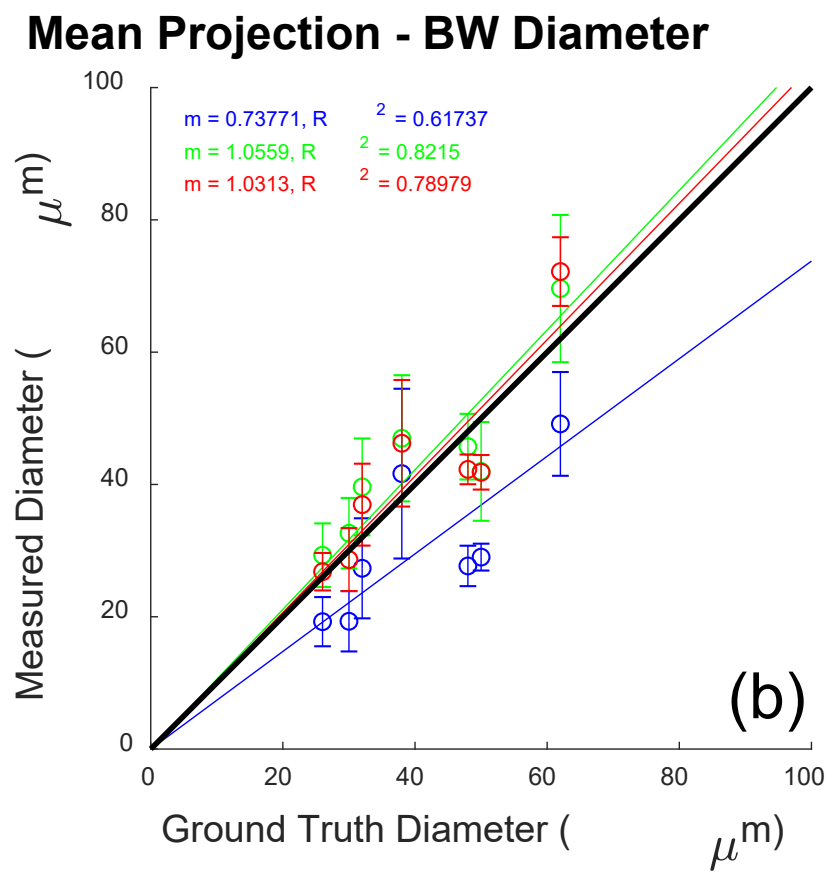

Max Projection - BW Diameter
m = 0.48343, R 2 = 0.53758
m = 0.86149, R 2 = 0.89032
m = 0.82089, R 2 = 0.68763
(b)
Measured Diameter (μm)
Ground Truth Diameter (μm)

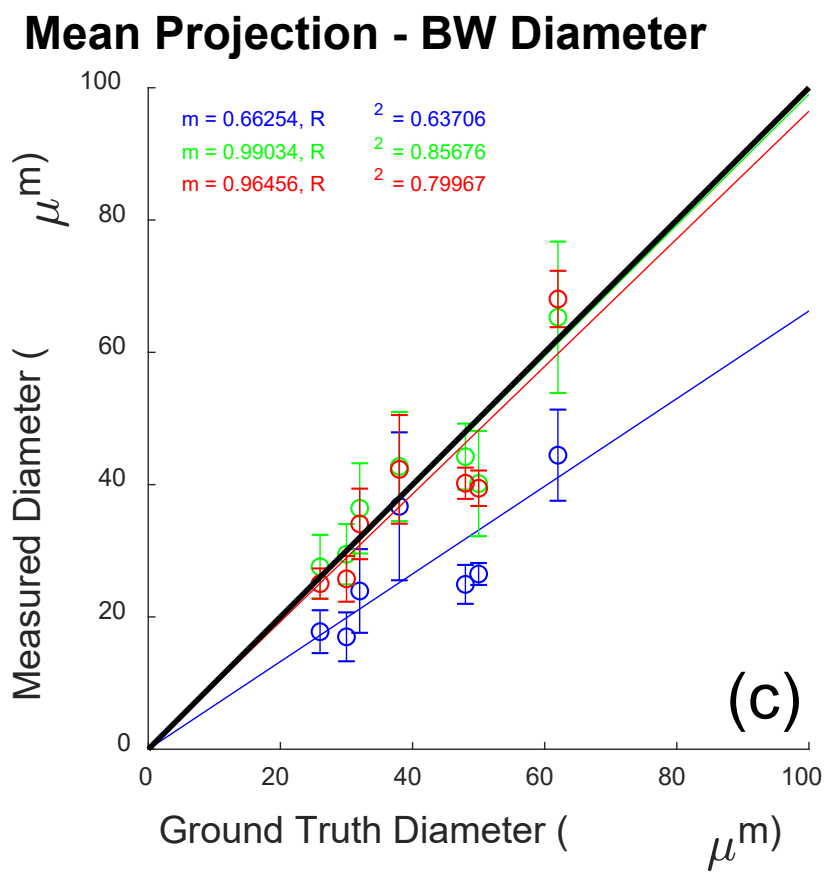

Max Projection - BW Diameter
m = 0.38089, R 2 = 0.44599
m = 0.784, R 2 = 0.86395
m = 0.72669, R 2 = 0.5741
(c)
Measured Diameter (μm)
Ground Truth Diameter (μm)

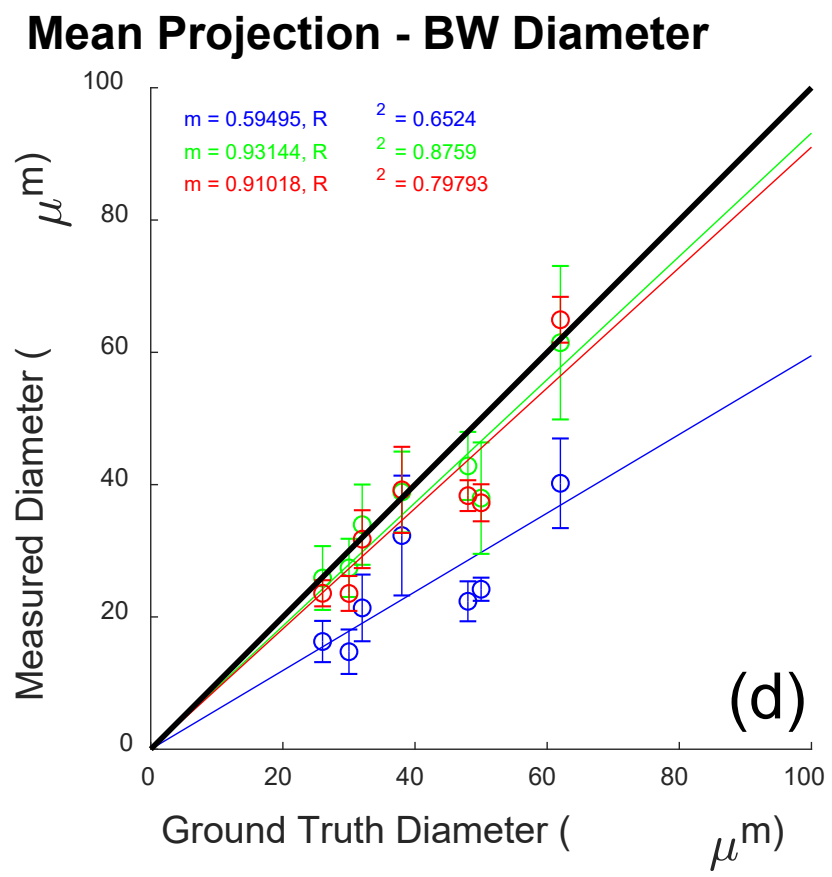

Max Projection - BW Diameter
m = 0.29961, R 2 = 0.34448
m = 0.71338, R 2 = 0.81248
m = 0.63247, R 2 = 0.45058
(d)
Measured Diameter (μm)
Ground Truth Diameter (μm)

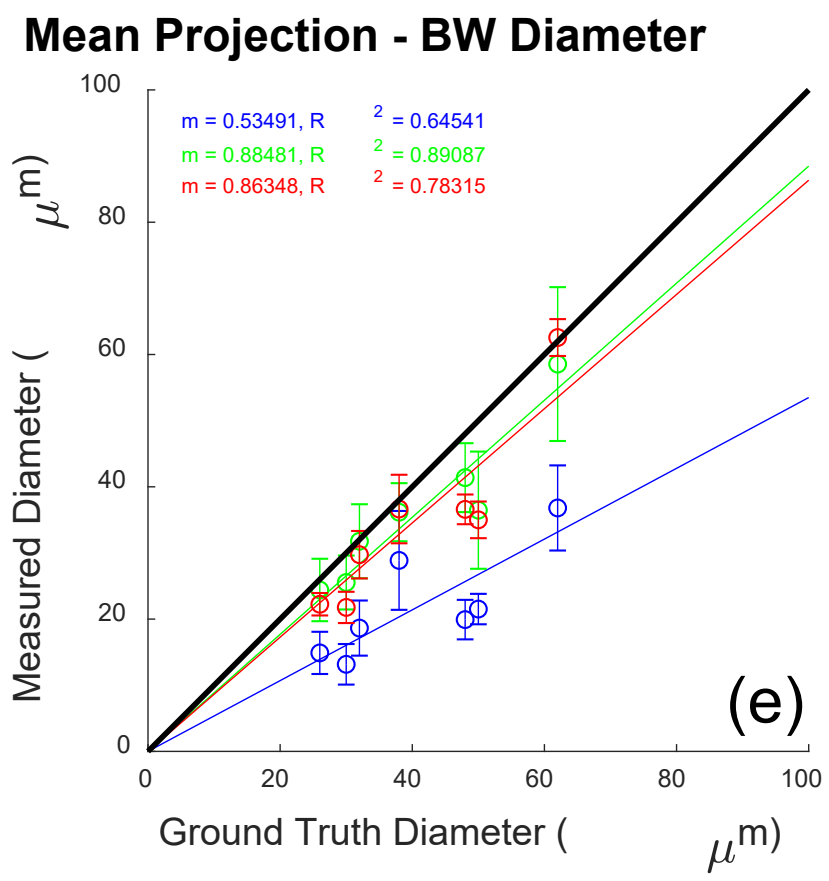

Max Projection - BW Diameter
m = 0.23893, R 2 = 0.24065
m = 0.64784, R 2 = 0.73451
m = 0.53885, R 2 = 0.34062
(e)
Measured Diameter (μm)
Ground Truth Diameter (μm)

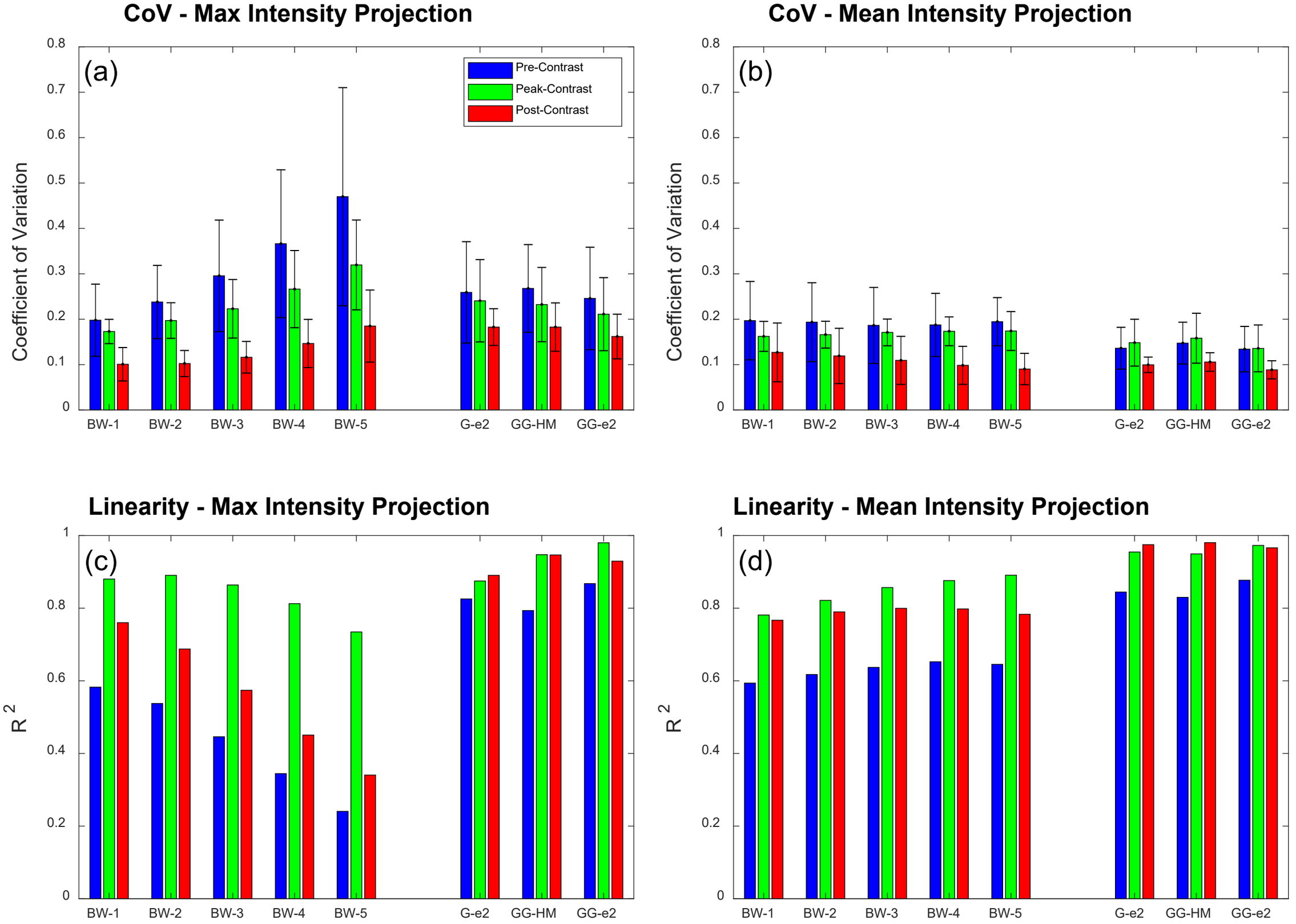

Mean Projection - BW Diameter
m = 0.83392, R 2 = 0.59383
m = 1.1371, R 2 = 0.7812
m = 1.1134, R 2 = 0.76683
(a)
Measured Diameter (μm)
Ground Truth Diameter (μm)

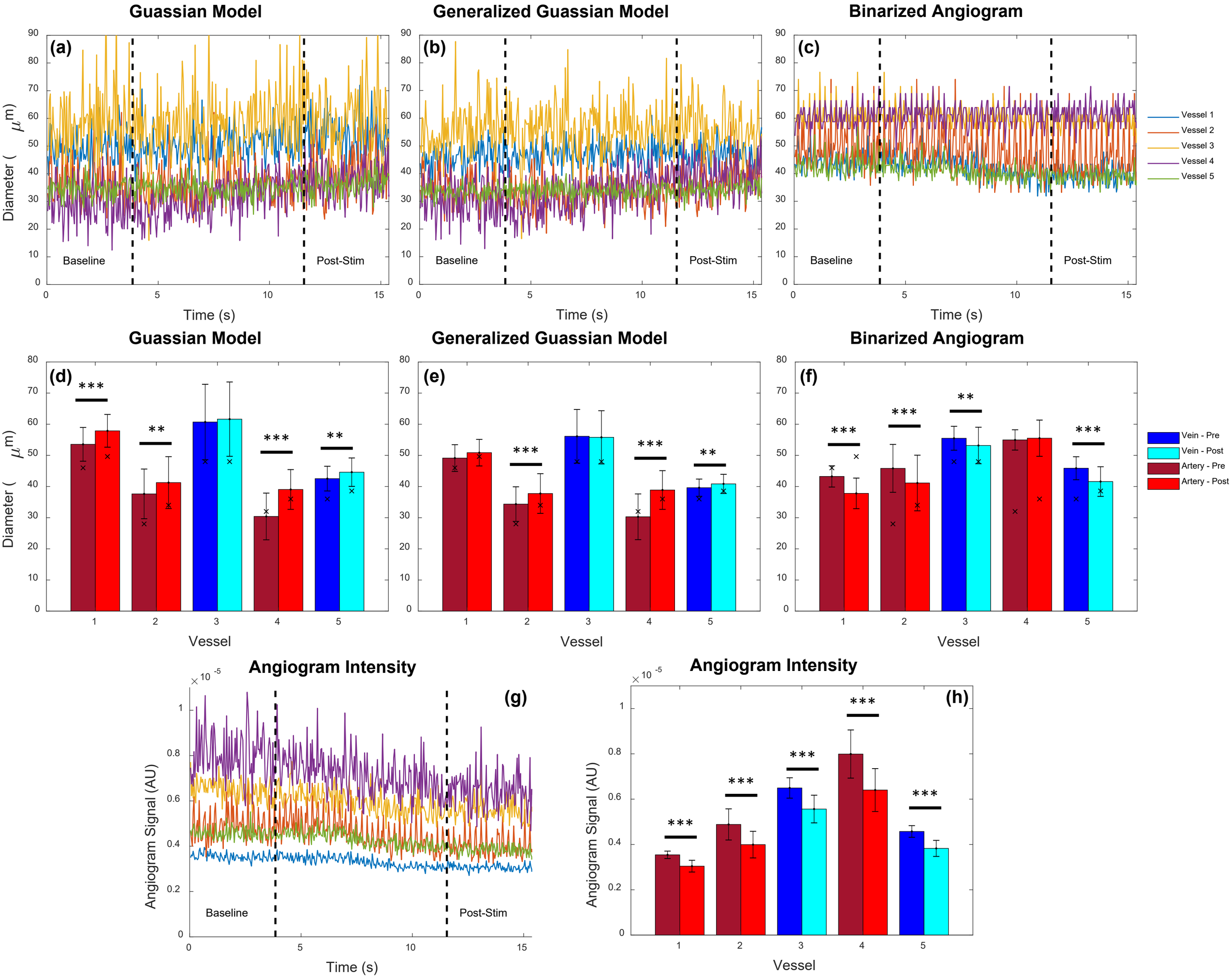

Mean Projection - BW Diameter
m = 0.73771, R 2 = 0.61737
m = 1.0559, R 2 = 0.8215
m = 1.0313, R 2 = 0.78979
(b)
Measured Diameter (μm)
Ground Truth Diameter (μm)

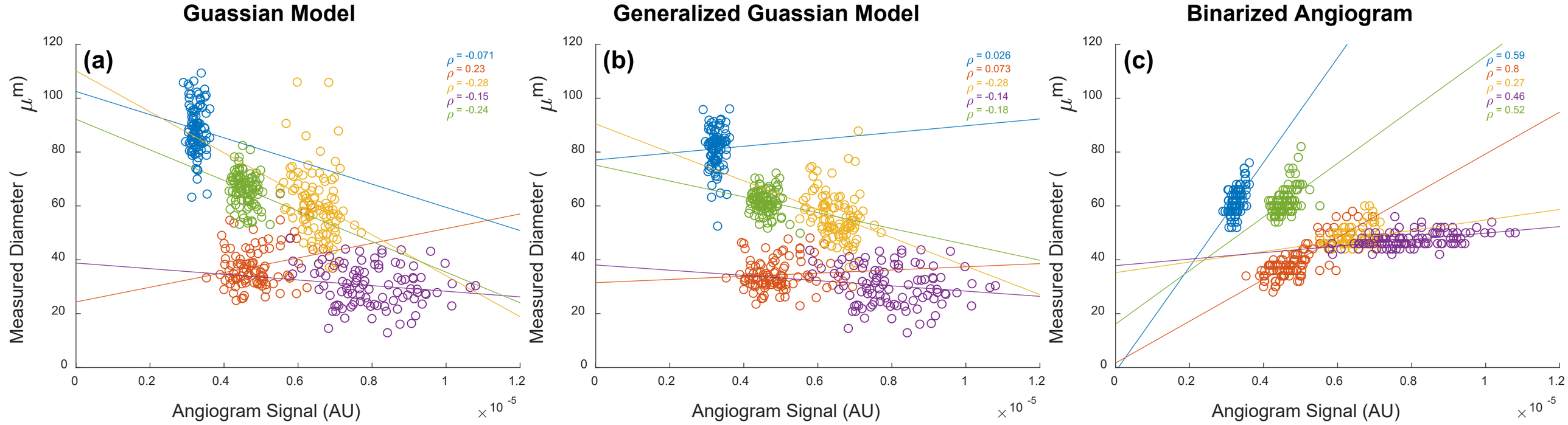

Mean Projection - BW Diameter
m = 0.66254, R 2 = 0.63706
m = 0.99034, R 2 = 0.85676
m = 0.96456, R 2 = 0.79967
(c)
Measured Diameter (μm)
Ground Truth Diameter (μm)


Mean Projection - BW Diameter
m = 0.59495, R 2 = 0.6524
m = 0.93144, R 2 = 0.8759
m = 0.91018, R 2 = 0.79793
(d)
Measured Diameter (μm)
Ground Truth Diameter (μm)


Mean Projection - BW Diameter
m = 0.53491, R 2 = 0.64541
m = 0.88481, R 2 = 0.89087
m = 0.86348, R 2 = 0.78315
(e)
Measured Diameter (μm)
Ground Truth Diameter (μm)


CoV - Max Intensity Projection
(a)
Pre-Contrast
Peak-Contrast
Post-Contrast
Coefficient of Variation
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
BW-1
BW-2
BW-3
BW-4
BW-5
G-e2
GG-HM
GG-e2
CoV - Mean Intensity Projection
(b)
Coefficient of Variation
Linearity - Max Intensity Projection
(c)
R 2
0
0.2
0.4
0.6
0.8
1
Linearity - Mean Intensity Projection
(d)
R 2


Guassian Model
Generalized Guassian Model
Binarized Angiogram
(a)
(b)
(c)
Diameter (μm)
Baseline
Post-Stim
Time (s)
Vessel 1
Vessel 2
Vessel 3
Vessel 4
Vessel 5
(d)
(e)
(f)
Vessel
Vein - Pre
Vein - Post
Artery - Pre
Artery - Post
Angiogram Intensity
(g)
(h)
Angiogram Signal (AU)
×10 -5


Guassian Model
Generalized Guassian Model
Binarized Angiogram
(a)
(b)
(c)
ρ = -0.071
ρ = 0.23
ρ = -0.28
ρ = -0.15
ρ = -0.24
ρ = 0.026
ρ = 0.073
ρ = -0.28
ρ = -0.14
ρ = -0.18
ρ = 0.59
ρ = 0.8
ρ = 0.27
ρ = 0.46
ρ = 0.52
Measured Diameter (μm)
Angiogram Signal (AU)
× 10 -5
0
20
40
60
80
100
120
0.2
0.4
0.6
0.8
1
1.2

Flicker Stimulation – Change in Diameter	**Vessel 1**		**Vessel 2**		**Vessel 3**		**Vessel 4**		**Vessel 5**	
Gaussian Model Diameter (% Change)	**9.2**	p < 0.0005	**9.6**	p = 0.0019	1.5	p = 0.59	**28.5**	p < 0.0005	**4.9**	p = 0.0008
Generalized Gaussian Model Diameter (% Change)	1.5	p = 0.28	**9.9**	p < 0.0005	-0.6	p = 0.79	**28.4**	p < 0.0005	**3.1**	p = 0.0034
Binarized Angiogram Diameter (% Change)	**-12.5**	p < 0.0005	**-10.2**	p < 0.0005	**-4.2**	p = 0.0011	1.0	p = 0.404	**-9.3**	p < 0.0005
Angiogram Intensity (% Change)	**-14.1**	p < 0.0005	**-18.2**	p < 0.0005	**-14.3**	p < 0.0005	**-19.9**	p < 0.0005	**-16.3**	p < 0.0005

Flicker Stimulation – Correlation to Intensity	**Vessel 1**		**Vessel 2**		**Vessel 3**		**Vessel 4**		**Vessel 5**	
Gaussian Model Diameter Correlation	-0.07	p = 0.48	**0.23**	p = 0.019	**-0.28**	p = 0.0044	-0.15	p = 0.14	**-0.24**	p = 0.018
Generalized Gaussian Model Diameter Correlation	0.03	p = 0.79	0.07	p = 0.47	**-0.28**	p = 0.0056	-0.14	p = 0.16	-0.18	p = 0.081
Binarized Angiogram Diameter Correlation	**0.59**	p < 0.0005	**0.80**	p < 0.0005	**0.27**	p = 0.0077	**0.46**	p < 0.0005	**0.52**	p < 0.0005

Overall correlation
Gaussian = -0.101
GG = -0.098
BW = 0.523